

When Satellite and Street Views Disagree:
Cross-View Complementarity in Disaster Damage Assessment

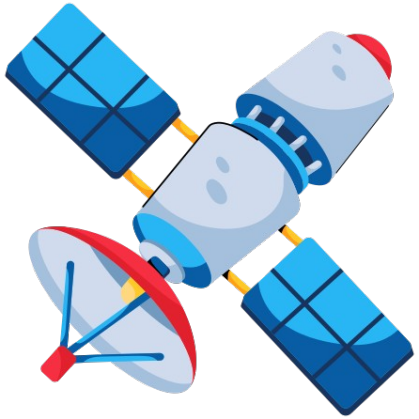
Yifan Yang, Ph.D. Student
Advisor: Dr. Lei Zou (lzou@tamu.edu)
Department of Geography, Texas A&M University
GEAR Lab Website: https://www.geoearlab.com

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Background

Satellite imagery: regional coverage within hours, but mostly roofs



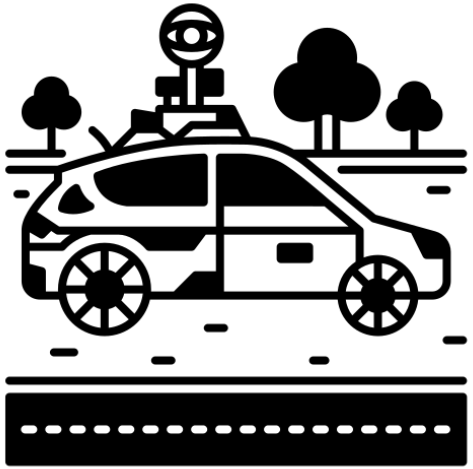


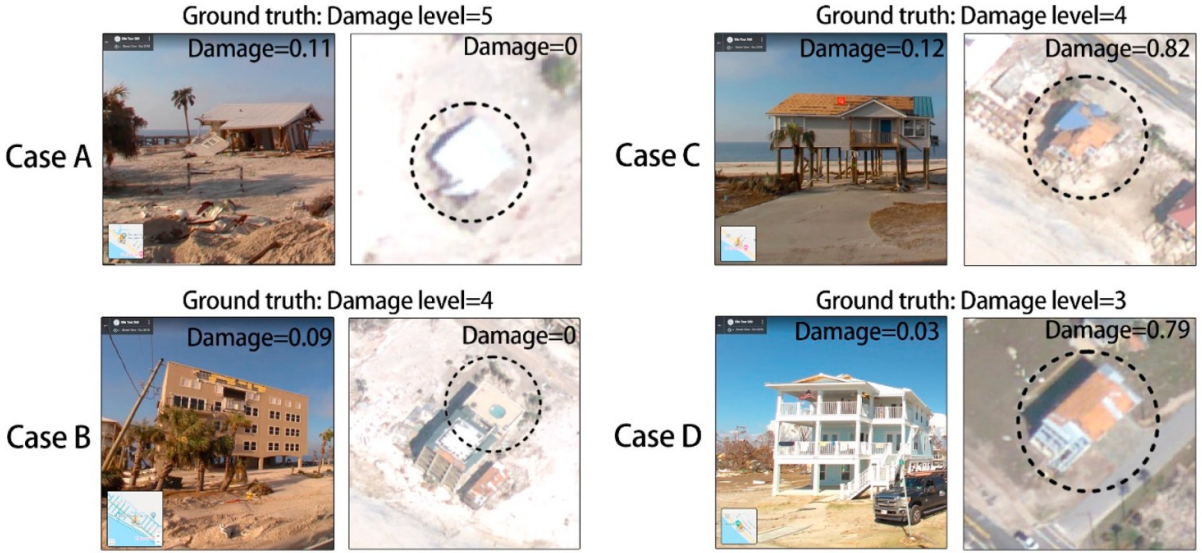
Strength: covers a whole region within hours; the basis of xBD-style mapping. **Limit:** an intact roof says little about the walls.

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Background

Street-view imagery: facades and debris that overhead sensors miss



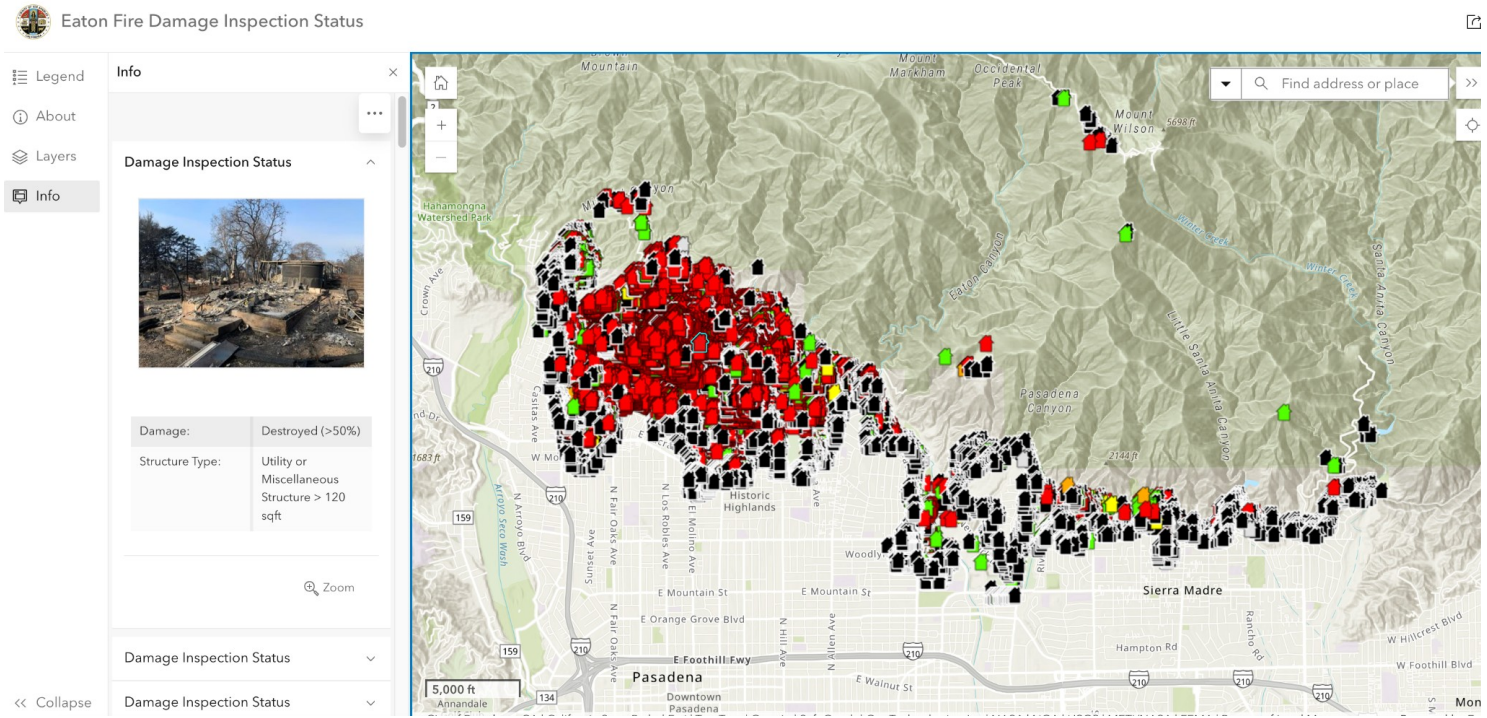


Same building, different answers: Case A is damage level 5 from the ground and reads as 0 from above.

Sometimes only one view sees the damage.

Background

Eaton Fire: ground photos aimed at one building each



Eaton wildfire, Altadena, CA (Jan 2025): CAL FIRE DINS building-level inspections. lacounty.maps.arcgis.com

Why it matters

Every structure visited, photographed, and labeled on a six-level scale.

Key word: aimed

The photo targets one building, paired with a satellite tile of the same parcel.

Problem & Research Questions

Average accuracy hides whether fusion uses both views



Most buildings look the same in both views. We mine the **conflict cases**, where independently trained single-view models disagree, and ask:

RQ1 How much complementary headroom does fusion leave on conflicts?

RQ2 Can a simple, readable gate recover part of it?

RQ3 Does aiming the ground view at the building change the value of fusion?

RQ4 Does the spatial density of conflicts track damage, without labels?



Data

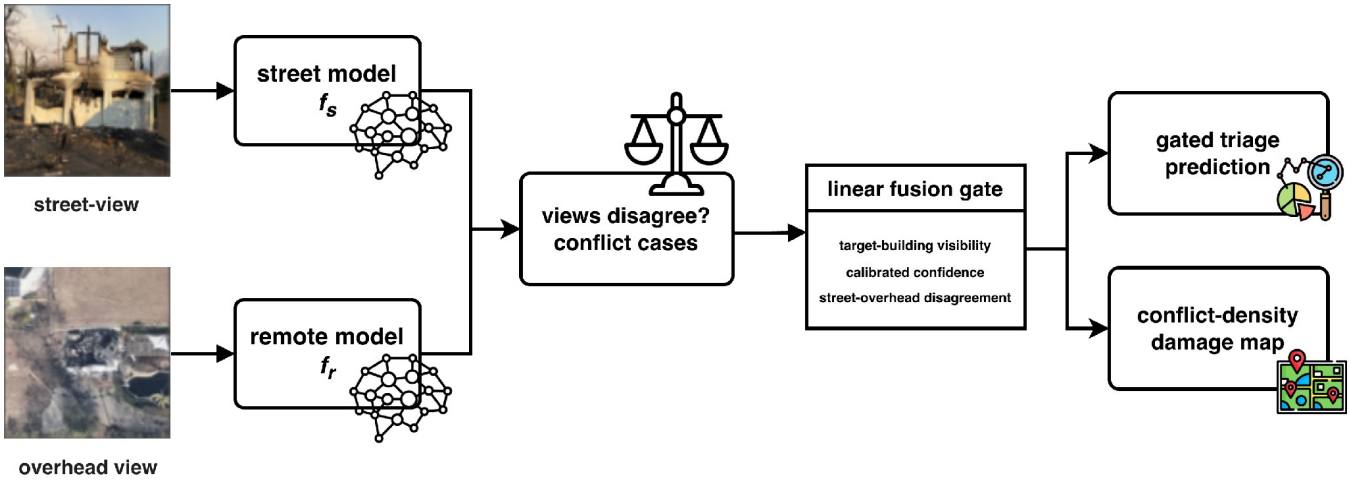
Two hazards × two ground-view regimes, three paired datasets

	Eaton wildfire (Altadena, CA)	Hurricane Ian (Sanibel Island, FL)	Hurricane Milton (Florida)
Event	January 2025 wildfire	September 2022 hurricane	October 2024 hurricane
Ground view	CAL FIRE DINS inspection photos, property-centric (aimed at a building)	Mapillary 360° panoramas (CVIAN pairing, CVDisaster)	360° street-view panoramas, built the same way
Overhead view	Post-event VHR tiles	27 cm post-event satellite imagery	Post-event satellite imagery
Labels	Six-level DINS scale → 3 ordinal classes (no/trace, damaged–repairable, destroyed)	Three-level damage perception	Three-level damage perception
Train / val / test pairs	6.5k / 1,966 / 1,984	3,248 / 573 / 300	1,740 / 307 / 254
Conflict rate (test)	**9.6%**	**33.2%**	**25.7%**

Two hazards × two ground-view regimes: photographs aimed at a building vs. panoramas that are not. Eaton is split by property and tile, so all photographs of one structure fall in one partition.

Method

Mine conflicts, measure headroom, gate, map disagreement



- **Conflicts:** street vs. remote models disagree
- **Oracle:** picks the correct view → headroom
- **Gate:** linear, 11 visibility, confidence, and disagreement features
- **FOV test:** building-centered vs. random 90° crops
- **Density:** conflicts per tile as a label-free damage map

Figure 1 of the paper. The implemented gate also mixes in the end-to-end fusion model (omitted in the diagram).

Method · Conflict Cases and the Oracle Gap

The oracle gap measures the evidence fusion leaves unused

- **Four models, one protocol, five seeds:** street_only, remote_only (ResNet-18), concat, and crossview (end-to-end fusion reference).
- **Conflict subset:** C = { x : s(x) ≠ r(x) }, the test samples where the two single-view predictions differ.
- **Oracle single view:** a conflict counts as correct if either view is correct, the headroom of hard view selection.
- **Calibration:** one temperature per view; training-free baselines use calibrated probabilities.



Conflicts with GT, street, remote, crossview predictions

Oracle-gap closure

closure(m) = [acc_C(m) − acc_C(best single)] / [acc_C(oracle) − acc_C(best single)]

0 = best single view, 1 = oracle

Method · Visibility-Conditioned Fusion Gate

A linear gate over 11 visibility and confidence cues

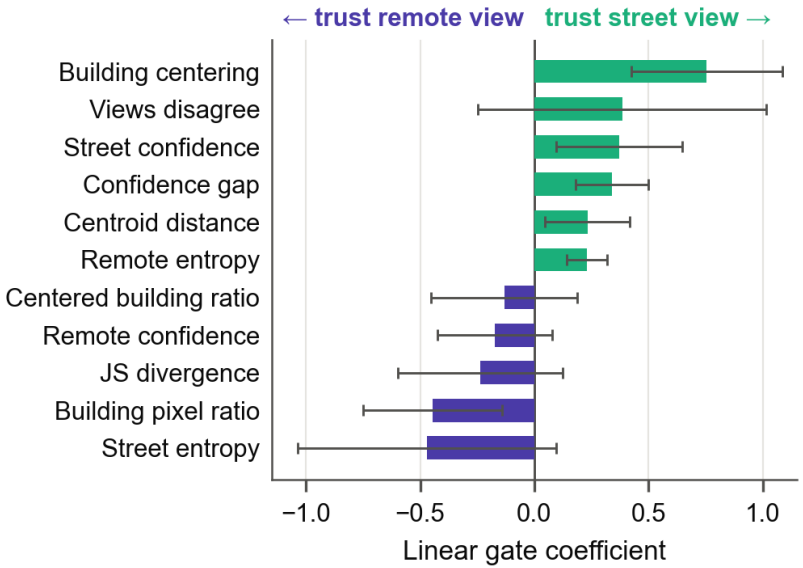
Eleven z-scored features per sample

- **Building visibility** (frozen SegFormer-B0): pixel ratio, centered-crop ratio, centering, centroid distance
- **Prediction cues:** calibrated confidence and entropy per view, confidence gap, JS divergence, disagreement flag

Mixture of experts with linear logits

$$p(y \mid x) = \Sigma_k\, w_k(z)\, p_k(y \mid x), \quad w(z) = \mathrm{softmax}(Wz + b)$$

$k \in \{\text{street, remote, crossview}\}$; fit on validation by mixture NLL. One frozen segmentation pass plus a linear fit. An MLP did not help.



Two-expert gate coefficients on Eaton (mean ± std, five seeds). Positive = weight toward the street view.

Results

Fusion helps on conflicts; calibrated averaging is a strong baseline

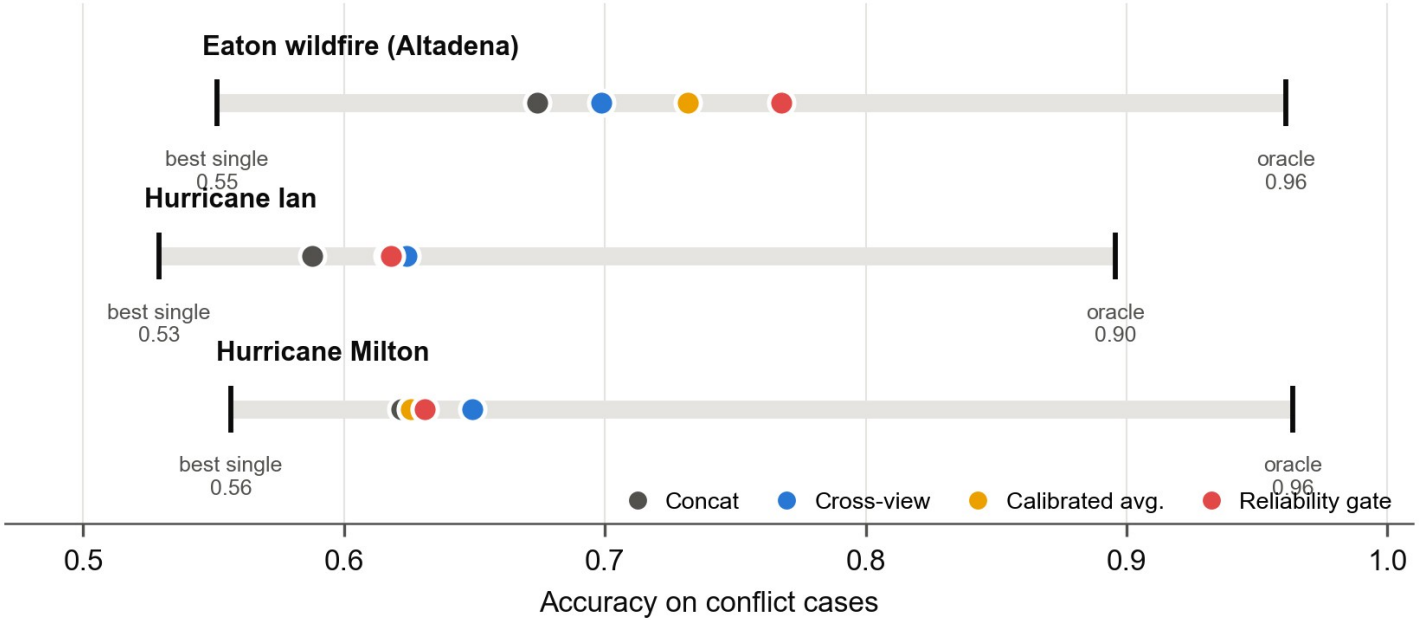
Test set, five-seed means (accuracy std ≤ 0.03). Conflict accuracy on each seed's conflict subset. Best non-oracle value per column in bold.

	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
Method	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.40	0.709	0.588	0.16	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.45	**0.735**	**0.624**	**0.26**	0.776	**0.649**	**0.23**
calibrated probability averaging	0.954	0.732	0.52	0.731	0.617	0.24	0.776	0.626	0.17
fusion gate (linear)	**0.959**	**0.768**	**0.59**	0.731	0.618	0.24	**0.777**	0.631	0.18
oracle single view	—	0.961	1.00	—	0.896	1.00	—	0.964	1.00

Fusion beats the street model on conflicts everywhere (Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). **Calibrated averaging is a strong baseline** once single-view models are converged (Eaton 0.954 vs. 0.940 overall). **The oracle dwarfs both:** +0.475 / 0.367 / 0.407 over the better single view and still +0.193 / 0.272 / 0.315 over the best fusion method.

RQ1 · Complementary Headroom

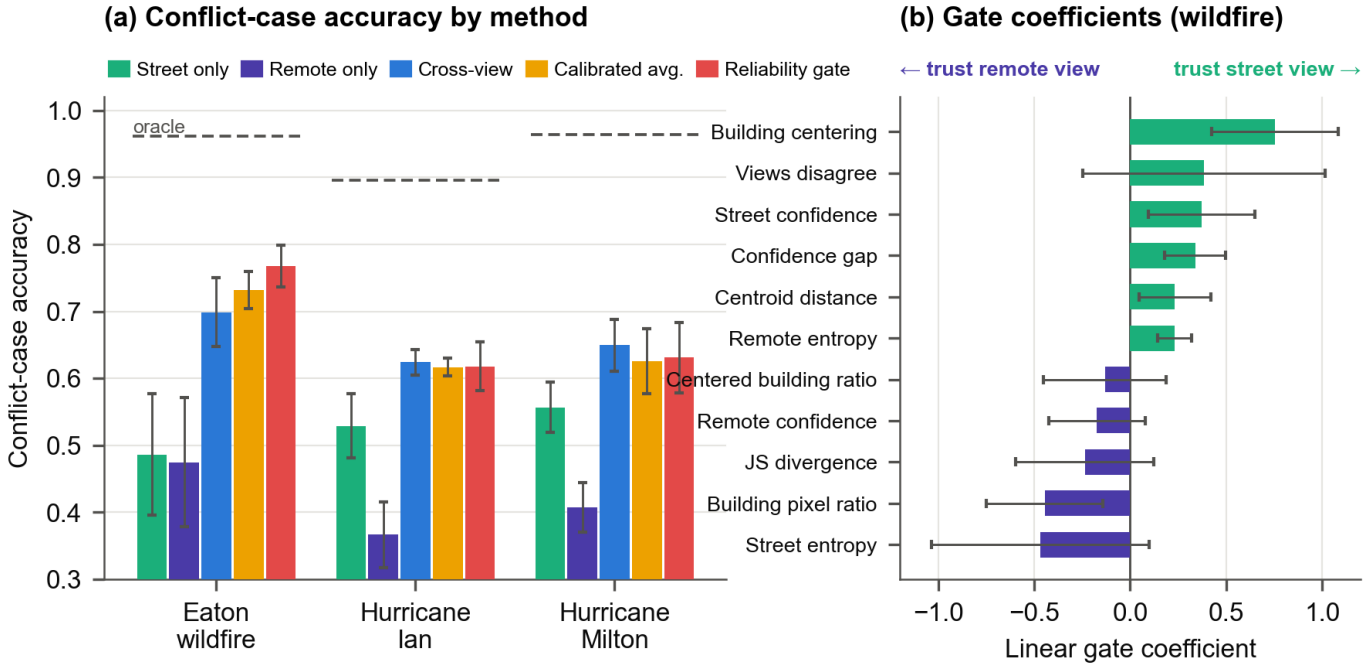
The oracle gap dwarfs every fusion gain



Each bar spans best single view → oracle on conflict cases; dots place each method. Picking the right single view per conflict would add 0.37–0.48 over the better single view; current fusion leaves most view-specific evidence unused.

RQ2 · Fusion Gate

The gate helps when photos are aimed at buildings



Eaton (aimed photos)

+0.072

conflict acc. vs. end-to-end fusion ($p < 10^{-4}$); +0.051 vs. averaging; closure 0.59

Ian / Milton (panoramas)

No significant gain: buildings fill 2.7% of pixels vs. 11.9% on Eaton

Two-expert gate alone (reweight street vs. remote only) reaches 0.734, on par with calibrated averaging and above end-to-end fusion (+0.034, p = 0.047); adding crossview as a third expert lifts it to 0.768. Largest coefficient: building centering (+0.75 ± 0.33); raw building pixel ratio is negative (−0.45 ± 0.31).

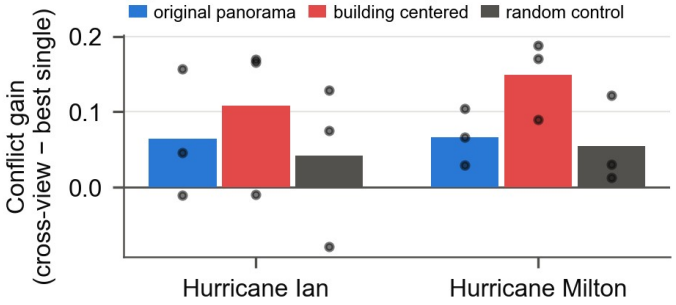
RQ3 · Field-of-View Intervention

Aiming the ground view at the building raises the value of fusion

(a) One panorama, two geometrically identical 90° crops



(b) Fusion benefit by crop variant





building crop



random crop

Same crop size, different pointing:
building azimuth vs. random azimuth

Conflict gain of fusion

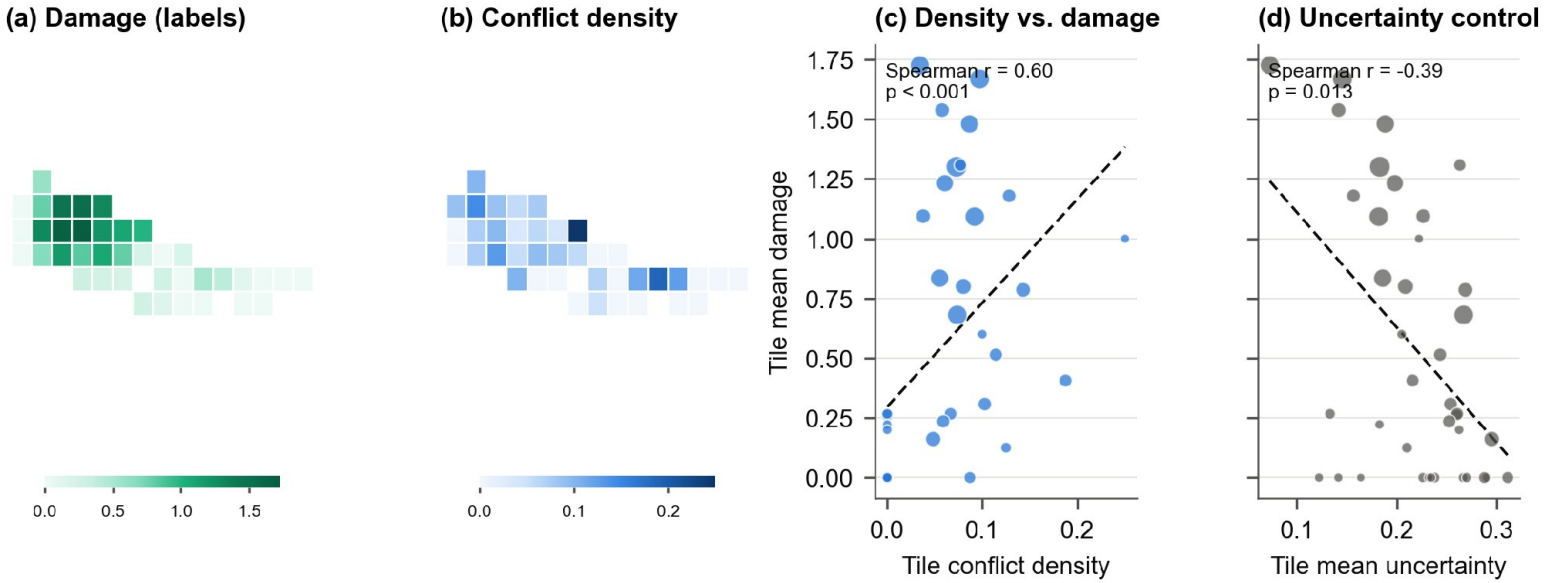
Ian 0.064 → **0.109** (random 0.042)
Milton 0.067 → **0.149** (random 0.055)

Building > random in 6/6 dataset–seed pairs. Street-only accuracy drops, so the gain is not a stronger street model.

Three seeds, three-epoch schedule; remote models unchanged. A fixed-sample test with target-building annotations would be stronger evidence.

RQ4 · Label-Free Damage Map

Conflict density tracks damage, with no labels (r = 0.615)



Eaton wildfire, 0.01° grid: tile conflict density (fraction of samples where the two argmax predictions disagree) tracks labeled damage, Spearman r = 0.615 (p = 0.001, 25 tiles), with no labels needed once the models are trained. **The uncertainty control anti-correlates** (r = −0.37 to −0.52): parcels burned to the ground are easy, confident classifications.

Caveats: damage is spatially autocorrelated (Moran's I = 0.27), so nominal p-values overstate the evidence; on Milton the signal is weaker and not significant (r = 0.289, p = 0.14) because better single-view models disagree less.

Qualitative Examples

In a conflict, one view sees the damage and the other does not



Wildfire, building-aimed (rows 1–3)

A gutted garage or facade under a roof that reads intact from above: the remote model says "no damage" with high confidence. In row 3, end-to-end fusion followed the overhead view and missed.

Hurricane, panoramic (rows 4–5)

The building is a small patch of the panorama; roadside debris drives the street prediction.

What a conflict is not

We measure prediction complementarity, not which component was damaged; occlusion, misalignment, or label ambiguity also cause conflicts.

Takeaways

Ask which view carries the evidence, where, and why

- **Conflict cases make complementarity measurable.** The oracle headroom, a restricted choice between two existing models, exceeds every fusion gain we observed: current fusion leaves most view-specific evidence unused.
- **Which view carries the evidence depends on hazard and capture setting.** On building-aimed wildfire photographs a cheap, auditable gate helps: weight the street view when it is confident and the building is centered. On hurricane panoramas calibrated averaging or end-to-end fusion suffices, and framing photographs on structures raises the value of fusion.
- **Spatially aggregated disagreement is a label-free damage indicator** (Eaton r = 0.615), and once single-view models are converged and calibrated, probability averaging is as strong as end-to-end fusion.
- *Mining conflicts turns "does fusion help?" into "which view carries the evidence, where, and why?"*

Code · data · paper

Yifan Yang and Lei Zou. 2026. When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment. In *The 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (GeoSearch '26)*, November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3849732.3857333

Code, protocols and results: https://github.com/rayford295/CrossViewGate · arXiv:2610.04327



Q&A

Welcome any comments/questions

yyang295@tamu.edu

github.com/rayford295/CrossViewGate